



Megan Barkdull

ASSISTANT CURATOR OF ENTOMOLOGY

Natural History Museum of Los Angeles County

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Professional appointments

Assistant Curator of Entomology

Natural History Museum of Los Angeles County

January 2025–present

Education

Ph.D, Department of Ecology and Evolutionary Biology

ADVISOR: DR. CORRIE MOREAU

Cornell University

August 2019–December 2024

Bachelor of Arts in Biology

ADVISOR: DR. EMILY HEFFERNAN

New College of Florida

August 2014–May 2018

- Honors Thesis: Population Genetics of the Threatened Florida Duskywing Skipper

Publications

Section sign (§) indicates co-senior or co-last authors. Double dagger (§) indicates co-first authors. Dagger (†) indicates a mentored undergraduate co-author.

Published:

1. **Barkdull, M.**, & C. S. Moreau. 2025. Phenotypic plasticity in turtle ants has opposing evolutionary consequences for genes and regulatory loci. *Evolution*. 10.1093/evolut/qpaf237
2. Donaldson, M.†, **Barkdull, M.**, & C. S. Moreau. 2024. Variation in aggression across ants (Hymenoptera: Formicidae) is linked to natural selection on neurotransmitter receptor and cuticular hydrocarbon synthesis genes. *Invited submission to Annals of the Entomological Society of America*. 10.1093/aesa/saae036
3. Heffernan, E. V.†, **Barkdull, M.†**, N. K. Brady. 2023. Microsatellites for butterfly conservation: historical challenges, current relevance, and a guide to implementation. *Frontiers in Ecology and Evolution*. 10.3389/fevo.2024.1344065
4. **Barkdull, M.†**, C. R. Miller†, M. A. Pipkin, & C. S. Moreau. 2023. Natural history collections: What are they and why do we need them?. *Frontiers for Young Minds*. <https://kids.frontiersin.org/articles/10.3389/frym.2023.1048608>
5. **Barkdull, M.**, & C. S. Moreau. 2023. Worker reproduction and caste polymorphism impact genome evolution and social genes across the ants. *Genome Biology and Evolution*. 10.1093/gbe/evad095
6. Heffernan, E., A. Markee., M. R. Truglio, **M. Barkdull**, S. Steele Cabrera, J. Daniels. 2023. Population Genetic Structure of a Rare Butterfly in a Fragmented South Florida Ecosystem. *Insects*. 10.3390/insects14040321
7. Williams, K.M., **M. Barkdull**, M. Fahmy, E. Hekkala, M. E. Siddall, S. Kvist. 2020. Caught red handed: iDNA points to wild source for CITES-protected contraband leeches. *Eur J Wildl Res*. 10.1007/s10344-020-01419-5
8. Siddall, M. E., **M. Barkdull**, M. Tessler, M. R. Brugler, E. Borda., & E. Hekkala. 2019. Ideating iDNA: Lessons and limitations from leeches in legacy collections. *PloS one*. 10.1371/journal.pone.0212226

In revision:

1. de Sá, F., Anderson, W., **Barkdull, M.**, Gaspari, G., Brasileiro, C., Taylor, S., Haddad, C., & K. Zamudio. Mating system dynamics in a Neotropical frog hybrid zone. In revision at *Evolution Letters*.
2. **Barkdull, M.**, & C. S. Moreau. Genome evolution in social insects. In revision at *Genome Biology and Evolution*; invited review submission.

In review:

1. Baker, Austin... **Barkdull, M.§**, Brown, B.§, L. M. Guzman§. DNA barcoding of more than one million insect specimens indicates that at least one third of California's insect biodiversity remains undiscovered. In review at *Ecography*.

Grants, Fellowships & Awards

Research grants

(Amount directly awarded: \$96,240; Additional amount managed: \$935,025)

- California Institute for Biodiversity— California Insect Barcoding grant. *Assumed management of grant upon joining NHMLAC (\$698,977); additional \$78,740 awarded to PI Barkdull in 2025.*
- National Science Foundation— Collaborative Research: Digitization TCN: DBI-2102006: Extending Anthophila research through image and trait digitization (Big-Bee). *Subaward with lead PI Dr. Katja Seltmann. Assumed management of grant upon joining NHMLAC (\$236,048)*
- Society for the Study of Evolution— Rosemary Grant Advanced Award (\$3,500)
- Cornell University— Andrew W. Mellon Student Research Grant (\$1,000)
- Sigma Xi, Cornell Chapter— Research Grant (\$1,000)
- American Society of Naturalists— Student Research Award (\$2,000)
- Society for Integrative and Comparative Biology— Grant-in-aid of research (\$1,000)
- Cornell University Graduate School— Research Travel Grant (\$2,000)
- Cornell University— Graduate Student Research Fund (\$4,000)
- Cornell University— Richard G. Harrison Fund (\$3,000)

Conference travel grants

(Total amount: \$1,250)

- Cornell University— Graduate School Conference Travel Grant (\$600)
- Cornell University— Graduate School Conference Travel Grant (\$500)
- Cornell University— Orenstein Fund (\$750)

Fellowships

(Total amount: \$244,487)

- National Science Foundation— Graduate Research Fellowship (\$138,000)
- Cornell University Insect Collection— Summer Stipend Award (\$10,832)
- Cornell Fellowship (\$33,241)
- National Merit Scholar (\$62,414)

Awards

- Book Award for Best First-Year Symposium Presentation

Talks

Dagger (†) indicates presenter.

Invited seminars:

1. Harvey Mudd College, Claremont, CA (2026). One genotype, many phenotypes: Evolutionary insights from complex traits in ants.
2. University of Southern California, Los Angeles (2026). One genotype, many phenotypes: Evolutionary insights from complex traits in ants.
3. California Institute of Technology, Los Angeles (2026). One genotype, many phenotypes: Evolutionary insights from complex traits in ants.
4. University of California, Los Angeles (2025). One genotype, many phenotypes: Evolutionary insights from complex traits in ants.
5. Natural History Museum of Los Angeles County (2024). Evolutionary insights from complex traits in ants.
6. Essig Brunch Seminar Series, University of Berkeley, Berkeley, CA. (2023). Genomic causes and consequences of worker caste polymorphism in the ants.

Oral presentations:

1. Entomology, Portland, OR. (2025). A. Baker†, B. Brown, **M. Barkdull**, M. Guzman. The California Insect Barcoding Initiative: a million specimens and counting.
2. Entomology, Portland, OR. (2025). T. McGlynn†, Y. Kamal, **M. Barkdull**, C. Bennett, M. Angel, A. Shultz. Insects demonstrate unparalleled capacity to reflect infrared solar radiation.
3. Entomology, Portland, OR. (2025). Y. Kamal†, M. Barrett, **M. Barkdull**, A. Shultz, T. McGlynn. Shining a light on ant thermoregulation: near-infrared reflectivity across latitudes.
4. Evolution, Athens, GA. (2025) **Barkdull, M†**. Turtle ant soldiers evolve by repurposing ancient genes.
5. Evolution, Montreal, QB. (2024) **Barkdull, M†**. and Moreau, C. One genotype, many phenotypes: Evolutionary insights from complex traits in ants.
6. Evolution, Albuquerque, NM. (2023) **Barkdull, M†**. and Moreau, C. Social traits modulate genome evolution in ants.
7. Cornell Department of Ecology and Evolutionary Biology Winter Symposium, Ithaca, NY. (2022). **Barkdull, M†**. and Moreau, C. Social traits shape genome evolution in ants.
8. Social Insects in the Northeast Region meeting, Brooklyn, NY. (2020). **Barkdull, M.†** and Moreau, C. Formicidae caste determination: single cell and functional genomic techniques. **Cancelled due to COVID19.

Posters:

1. Entomology, Portland, OR. (2025). C. Bennett†, T. McGlynn, M. Barrett, A. Shultz, **M. Barkdull**, C. Penick. Ant near-infrared reflectivity responds to climatic conditions.
2. Biology of Genomes meeting, Cold Spring Harbor Laboratory, NY. (2022). **Barkdull, M†**. and Moreau, C. Social traits modulate genome evolution in ants.
3. International Symposium on Molecular Insect Science, Barcelona, Spain. (2019). E.V Heffernan†, **M. Barkdull**, and A. Markee. The role of gene flow in connecting populations of endangered skipper butterflies in the United States: using genetic data to inform management strategies.
4. Entomological Society of America meeting, St. Louis, MO. (2019). Markee, A.N†., Saarinen, E. V., **Barkdull, M.** Conservation genetics of the Florida duskywing skipper (*Ephyriades brunnea*): a multi-population assessment of a rare South Florida butterfly.

5. Imperiled Butterflies Working Group meeting, Miami, FL. (2017). Ash, M.†, **Barkdull, M.†**, Elmir, G.† Are endangered Dakota Skipper populations infected with Wolbachia?

Research Positions

Ph.D. researcher

Cornell University

MOREAU LAB

August 2018–December 2024

- Investigated the genetic basis and consequences of worker caste polymorphisms using genome sequencing, comparative genomic methods, and transcriptomics.
- Conducted multiple field excursions to collect ant specimens to support research program.

Post-baccalaureate intern

Cornell University

ZAMUDIO LAB

September 2018–August 2019

- Conducted conservation genetics research on a number of reptile and amphibian species.
- Responsible for assisting all lab members with wet lab work, including DNA extractions, microsatellite sequencing, Sanger sequencing, and library prep for MiSeq.
- Collaborated with Dr. Fábio de Sá to describe the genetic mating system of the saxicolous Brazilian frog *Cycloramphus boraceiensis*.

Hawkmoth Sensory Behavior volunteer

Cornell University

RAGUSO LAB

September 2018–December 2018

- Participated in weekly hawkmoth colony care tasks (feeding, plant care, pupae counts etc.).
- Ran experiments testing the role of floral humidity in hawkmoth feeding choice.

Birdsong Evolution, research assistant

Cornell Lab of Ornithology

DRS. ARAYA-SALAS AND ODOM

May 2018–May 2019

- Coded individual song elements in birdsong spectrograms for downstream data analysis.
- Created analysis protocol and generate graphics for future researchers on the project.

Florida Duskywing Genetics Project, research assistant

New College of Florida

HEFFERNAN LAB

August 2017–May 2018

- Carried out a population genetics analysis of the threatened Florida Duskywing skipper butterfly to inform conservation policy.
- Performed next-generation sequencing in order to develop novel microsatellite markers

NSF Research Experiences for Undergraduates Intern

American Museum of Natural History

DEPARTMENT OF INVERTEBRATE ZOOLOGY

May 2017–August 2017

- Used genetic techniques (Sanger sequencing etc.) to identify the hosts of terrestrial bloodfeeding leeches.
- Conducted fieldwork to collect invertebrate (leeches, crayfish) for related lab projects.

Dakota Skipper Wolbachia Status, research assistant

New College of Florida

HEFFERNAN LAB

January 2017–May 2017

- Conducted DNA extractions and PCR amplifications of Wolbachia pathogen genes from the endangered Dakota Skipper butterfly to identify pathogen strains.

Curatorial Skills & Experiences

From Nabokov's Net exhibit contributor

CORNELL UNIVERSITY INSECT COLLECTION AND MANN LIBRARY

September 2023–March 2024

- Collaborated with CUIC and Mann library staff to design an exhibit featuring the butterfly specimens collected by Vladimir Nabokov during his time at Cornell.
- Designed visuals and wrote content for exhibit component focused on butterfly conservation and the role of natural history collections in conservation research.

Six-Legged Science: Unlocking the Secrets of the Insect World exhibit contributor

CORNELL UNIVERSITY INSECT COLLECTION AND MUSEUM OF THE EARTH

January 2022–March 2022

- Helped design components of public-facing museum exhibit, including pulling specimens, databasing, and design work
- This exhibit reached 18,592 in-person visitors.

Social Media co-manager

CORNELL UNIVERSITY INSECT COLLECTION

January 2022–December 2024

- Co-managed the CUIC's Instagram and Twitter pages, designing and posting content to engage the broader community with the work of the CUIC.

Collections Management Workshop, participant

CORNELL UNIVERSITY INSECT COLLECTION

January 2021–May 2021

- Participant in a semester-long workshop series, led by Drs. Jason Dombroskie and Corrie Moreau, which focused on practical discussion of topics related to the care and management of natural history collections, including museum mission statements, managing loans, permits, managing volunteers and students, databasing, and outreach.

Collection Databasing, graduate supervisor

CORNELL UNIVERSITY INSECT COLLECTION

August 2020–December 2024

- Responsible for databasing specimens in the CUIC's Formicidae collection, from data entry to interfacing with Specify.
- Hired, trained, and supervised undergraduate and graduate students on how to conduct databasing in the the collection.
- Designed training materials for other databasing workers.

Entomological Collections Network annual meeting, attendee

November 2020–NA

- Virtually attended conferences sessions on a wide range of entomological collections management topics.

Teaching

Educator Workshop facilitator

Natural History Museum of Los Angeles County

May 2026–NA

- Co-taught a workshop for K12 educators focused on ant and spider biology, and on using iNaturalist in classroom settings.

Guest lectures and collections tours

Various

VARIOUS

January 2025–ongoing

- Guest lectures and tours of the Entomology Collection for courses at Loyola Marymount University (x2), Harvey-Mudd College (x1), and Victor Valley College (x1).

Introduction to Evolution and Diversity teaching assistant

BIOEE1780

Cornell University

January 2020–December 2024

- Taught three weekly discussion/lab sections.
- Assisted with course transition to an online format in 2020 due to COVID-19.

Introduction to Evolution and Diversity Writing in the Major guest lecturer

Cornell University

BIOEE 1780

October 2023–NA

- Invited to give a guest lecture on my dissertation research and path to graduate school in Jasmine Peters' Writing in the Majors course.

Conservation Biology guest lecturer

Forham University

BISC 4575

April 2023–NA

- Invited to give a guest lecture on conservation genetics and my research to Dr. Evon Hekkala's conservation biology seminar course.

Introduction to Evolution and Diversity grader

Cornell University

BIOEE 1781

August 2022–December 2023

- Responsible for grading weekly assignments for online evolution and diversity course.

Comparative Physiology teaching assistant

Cornell University

BIOG1440

August 2019–December 2020

- Taught three weekly discussion/lab sections, leading hands-on investigations of physiological concepts covered in lectures.
- Created extensive, novel course review materials; available on my GitHub.

Foundations of Biology teaching assistant

New College of Florida

BIOL2100

August 2017–December 2017

- Led review sessions prior to tests and quizzes.
- Met individually with students to address performance questions.

Cellular Biology teaching assistant

New College of Florida

BIOL3500

August 2016–December 2016

- Led content-delivery and problem-solving review sessions once per week.
- Assisted in exam grading and clerical tasks.

Mentoring

Austin Baker

POSTDOCTORAL RESEARCHER

NHM

January 2025–November 2025

- Austin was the lead postdoc on the California Insect Barcoding Initiative, a statewide collaborative project to collect and DNA barcode every insect species in California.

Yume Kamal

MASTER'S STUDENT

NHM/CSU Dominguez Hills

March 2025–NA

- With Dr. Terry McGlynn, co-mentored on a master's project to understand how variation in ants' infrared reflectivity is associated with thermal environment.

Eduardo Guzman

HIGH SCHOOL SUMMER INTERN

Cornell University

June 2024–NA

- Co-mentored Eduardo as he gained exposure to a variety of entomological and collections-based research approaches, including field collecting, ant specimen curation, databasing, DNA extraction, PCR, and scientific communication.

Mahalia Donaldson

PAID RESEARCH INTERN

Cornell University

August 2022–May 2023

- Conducted a research project to the evolution of genes related to aggression in ants spanning a range of aggression phenotypes.
- Learning scientific programming skills including Bash and R scripting, Git and Github version control
- Results of this project were published as an invited submission to Annals of the Entomological Society of America.

Seyde Delgado

PAID RESEARCH INTERN; BIOLOGY 2990 INDEPENDENT RESEARCH EXPERIENCE STUDENT

Cornell University

May 2022–December 2024

- Conducted a research project to understand the relationship between herbivory and molecular evolution in ants.
- Learning scientific programming skills including Bash and R scripting, Git and Github version control

Service

NSF reviewer

NATIONAL SCIENCE FOUNDATION

December 2025–NA

SSE Lewontin Early Award reviewer

SOCIETY FOR THE STUDY OF EVOLUTION

March 2025–NA

Professional Development panelist

DEPT. OF ENTOMOLOGY

Cornell University

October 2024–NA

- Answered questions from first-year graduate students about strategies for success in graduate school.

EvoAlly

SAFE EVOLUTION INITIATIVE

Evolution Conference

July 2024–Present

- Trained to use active bystander techniques and serve as a point of contact for conference attendees to increase participant safety as part of the Safe Evolution initiative.

Graduate Student Association treasurer

DEPT. OF ECOLOGY & EVOLUTIONARY BIOLOGY

Cornell University

August 2022–August 2023

- Managed funding from the university for graduate student and some department events.

Graduate Student Association co-president

DEPT. OF ECOLOGY & EVOLUTIONARY BIOLOGY

Cornell University

July 2020–August 2022

- Liased between graduate student body and faculty leadership to communicate and address graduate student concerns
- Participated in the formation of working groups focused on assessing and improving diversity, equity and inclusion in the department.

Cornell Diversity Preview Weekend volunteer

DEPT. OF ECOLOGY & EVOLUTIONARY BIOLOGY

Cornell University

September 2020 and summer 2021–NA

- Volunteered for the workshops "Crafting a CV" and "Fellowships and Personal Statements" providing peer review to participants for the 2020 DPW event.
- Served as a peer mentor to a DPW participant during the Fall 2020 semester.
- Answered participant questions as a panelist for the workshop "Crafting a CV" for the 2021 DPW event.

EvoDay Planning Committee

* POSTPONED TO SPRING 2022 DUE TO COVID19

Cornell University

Spring 2020–NA

- Identify and invite speakers for a day-long symposium on the theme of "Evolution in Deep Time".

Department Seminar Series Committee

DEPT. OF ECOLOGY & EVOLUTIONARY BIOLOGY

Cornell University

August 2019–August 2020

- This committee plans and coordinates the department's weekly seminar series
- Responsible for soliciting speaker nominations, constructing schedule of speakers, and assisting host labs with managing their speaker visits.
- While on this committee, I initiated a push to increase the diversity of our invited speakers, based on the best practices identified by Hagan et al. (2020).

Journal referee:

- Integrative and Comparative Biology, editorial reviewer
- Environmental DNA, ad hoc reviewer
- Wellcome Open Research, ad hoc reviewer
- Molecular Biology and Evolution, ad hoc reviewer
- Ecology and Evolution, ad hoc reviewer
- PCIEvolBio, ad hoc reviewer
- Molecular Ecology, ad hoc reviewer
- Genome Biology and Evolution, ad hoc reviewer
- Evolution, ad hoc reviewer
- Insectes Sociaux, ad hoc reviewer

Society membership:

- Society for the Study of Evolution (SSE), 2023- present.
- Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology (SORTEE), 2023- present.
- Society for Integrative and Comparative Biology (SICB), 2022- 2023.
- Entomological Society of America (ESA), 2024- present.

Outreach

Tabling, Coyotl wan Macehualli Lending Library

PRESENTER

NHMLAC

June 2026–NA

- Tabled with ant and arachnid specimens to highly ongoing community science projects in the Entomology department.

Tabling, NHMLAC Bug Fair

PRESENTER

NHMLAC

May 2025, May 2026–NA

- Manned a table on insect diversity, bringing our collection to members of the public.

Collections tours

FACILITATOR

NHMLAC

May 2026–NA

- Led behind-the-scenes tours of the NHM Entomology collection for first grade students, teachers, and parent chaperones.

Student iNaturalist walks

FACILITATOR

NHMLAC and Anawakalmekak School

April 2026–NA

- As part of Anawakalmekak's Earth Day Festival, facilitated iNaturalist walks for students of a variety of grade levels.
- Tabled with ant and arachnid specimens to highly ongoing community science projects in the Entomology department.

City Nature Challenge UV-Light Habitat Survey

FACILITATOR

Santa Monica Mountains Fund and the National Park Service

April 2026–NA

- Guided guests through an evening of blacklighting and contributing observations to iNaturalist as part of the 2026 City Nature Challenge.

The Midnight Mission GED science class and museum tour

PRESENTER

The Midnight Mission

April 2026–May 2026

- Co-led a class on arthropod diversity and a tour of NHMLAC for residents at The Midnight Mission.

Tabling, NHMLAC Earth Day Festival

EVENT VOLUNTEER

NHMLAC

April 2025–April 2026

- Manned a table on insect diversity, bringing our collection to members of the public.

Wilson High School field trip

PRESENTER

NHMLAC

March 2026–NA

- Presenter and activity leader for students during a field trip to Elephant Hill, focused on identifying ants and making contributions to iNaturalist.

Anawakalmekak Teacher Workshop

PRESENTER

NHMLAC

March 2026–NA

- Co-led a workshop focused on using iNaturalist and identifying ants in Los Angeles.

Senior Morning

PRESENTER

NHMLAC

March 2026–NA

- Gave local seniors a tour of the NHM Nature Gardens, focusing on insect diversity, and shared specimens from our collection.

Haunted Museum

PRESENTER

NHMLAC

September 2025–NA

- Manned a table on insect diversity, bringing our collection to members of the public.

Descanso After Dark

PRESENTER

Descanso Gardens

September 2025–NA

- Displayed live insects and presented on insect diversity to guests at the Descanso After Dark event.

Tours, NHMLAC Entomology

PRESENTER

NHMLAC

Ongoing–NA

- Gave tours of the NHMLAC Entomology collection to groups including the NSF RaMP UNLAB mentees, and in partnership with the Advancement team.

Tabling, NHMLAC Green Screens showing of the film Nocturnes

EVENT VOLUNTEER

NHMLAC

April 2025–NA

- Manned a table on moth diversity, bringing our collection to members of the public.

Interview, KTLA

March 2025–NA

- Discussed the NHMLAC Butterfly Pavilion and butterfly conservation with KTLA's Gayle Anderson.

Tabling, NHMLAC First Fridays

EVENT VOLUNTEER

NHMLAC

March 2025–NA

- Manned a table on insects and sound, bringing our collection to members of the public.

Guest presenter for second grade class unit on arthropods

VOLUNTEER

Ferndale Elementary School

January 2024–NA

- Presented ant fun facts to 60 second grade students, and answered student questions about ants.

Volunteer instructor for Ithaca Children's Garden summer camp

EVENT VOLUNTEER

Ithaca Children's Garden

July 2023–NA

- Led 6-10 year old campers through hands-on activity focused on the ant life cycle and on insect collecting.

Panelist for "Women in Entomology: Meet the Superheroes of Insect Science"

PUBLIC TALK

WSKG/PBS

October 2022–NA

- Discussed research and insect conservation.
- Panel can be viewed at <https://www.youtube.com/watch?v=HZ0WdZk84wM>

Guest blogger for the Paleontological Research Institute/Museum of the Earth

GUEST BLOGGER

Paleontological Research Institute/Museum of the Earth

July 2022–NA

- Guest blogged for the Paleontological Research Institute and Museum of the Earth. **Barkdull, M.** (2022). Insects under threat: the role of natural history collections in biodiversity conservation. <https://bit.ly/3ph2lsm>

"Meet an Entomologist" volunteer

EVENT VOLUNTEER

Six-legged Science Exhibit, Museum of the Earth

July 2022–NA

- Answered questions from visitors to the Museum of the Earth's Six-legged Science exhibit.

Panelist for "Exploring Insect Evolution: A Panel on New Research"

PUBLIC TALK

Darwin Days Week, Museum of the Earth

February 2022–NA

- Gave a research talk for the event *Exploring Insect Evolution: A Panel on New Research* as part of the Museum of the Earth's Darwin Days events.
- Talk can be viewed at <https://bit.ly/3s4jOGB>

Cornell Insectapalooza volunteer

EVENT VOLUNTEER

Dept. of Entomology, Cornell University

October 2019, 2022, 2023, 2024–NA

- Manned a table on ant diversity and discussed all things ant with members of the public, including young children, teens and adults.

